AWS | EC2 | APACHE HTTP SERVER

Deploy a Custom Apache Web Server on AWS EC2

Step-by-step project documentation | Region: US East (N. Virginia)

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AWS Region
US East (N. Virginia)
`us-east-1`

1. Project Overview

Goal: Launch an Amazon EC2 instance running Amazon Linux 2023 in the **US East (N. Virginia)** Region, install Apache HTTP Server, publish a custom `index.html` page using EC2 User Data (or by editing the file manually), and test it from a browser.

> **Note:** If the browser shows the default "It works!" page, Apache is running, but the custom `index.html` has not replaced the default page. Section 7 explains the fix. The website's HTML was generated with GPT; this document uses a short sample page as a placeholder.

2. Before You Start

- Sign in to the AWS Management Console: console.aws.amazon.com
- Select the Region **US East (N. Virginia)**, `us-east-1`, in the top-right Region selector.
- Check current AWS pricing and Free Tier eligibility for your account and instance type.
- HTTP port 80 is required for a public demo. SSH port 22 is optional and should be limited to your own IP.

3. Part A: Launch the EC2 Instance

1. **Open EC2:** Search for EC2 in the console, open **Instances**, then click **Launch instances**. Confirm the Region shows N. Virginia.
2. **Name:** Enter `Custom-Apache-Web-Server`.
3. **AMI:** Choose **Amazon Linux 2023**.
4. **Instance type:** Choose a small type such as `t2.micro` or `t3.micro`. Confirm pricing and eligibility.
5. **Key pair:** Optional for a web-only demo, but SSH will not work without one. A key pair or Systems Manager helps with troubleshooting.
6. **Network:** Enable **Auto-assign public IP** so the site opens via the public IPv4 address.
7. **Security group:** Create `Apache-Web-Server-SG` with inbound HTTP, TCP 80, source `0.0.0.0/0`. Optionally add SSH, TCP 22, source **My IP**.
8. **User Data:** Expand **Advanced details** and paste the script from Part B. Do not include Markdown fences.
9. **Launch:** Click **Launch instance**. Wait until the instance is Running and status checks pass, then allow a few minutes for User Data to finish.

4. Part B: User Data Script

Paste this into **Advanced details > User data**. It installs Apache and writes the web page. The HTML block is a placeholder.

```
#!/bin/bash
set -euxo pipefail

dnf update -y
dnf install -y httpd
mkdir -p /var/www/html

cat > /var/www/html/index.html <<'HTML'
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My Custom Apache Page</title>
</head>
<body>
  <h1>Hello from Apache on AWS EC2!</h1>
  <!-- Replace with your GPT-generated HTML code -->
</body>
</html>
HTML

chown -R apache:apache /var/www/html
chmod 755 /var/www/html
chmod 644 /var/www/html/index.html
systemctl enable --now httpd
```

Using GPT for the page: ask GPT for a complete single-file `index.html` (HTML, CSS and JavaScript together) and paste it between the `cat` line and the closing `HTML` line, without Markdown fences.

5. Part C: Verify the Website

1. **Find the public IP:** EC2 > Instances > select your instance > copy the Public IPv4 address.
2. **Open the site:** Go to `http://YOUR-PUBLIC-IP`. Use http, not https, because TLS is not configured.
3. **Wait:** Give cloud-init a few minutes to install packages and write the file.
4. **Refresh:** If you saw "It works!" earlier, press Ctrl+F5 or use an incognito tab.

6. Part D: Troubleshooting Commands

Connect with SSH or EC2 Instance Connect, then run:

Purpose	Command
Check the file exists	`sudo ls -l /var/www/html/`
Check the first lines	`sudo head -10 /var/www/html/index.html`
Check Apache service	`sudo systemctl status httpd --no-pager`
Test locally	`curl -s http://localhost | head -20`
Look for default text	`sudo grep -Ril 'It works!' /var/www/html /etc/httpd/conf.d`
Check document root	`sudo grep -R 'DocumentRoot' /etc/httpd/conf /etc/httpd/conf.d`
Review User Data log	`sudo tail -100 /var/log/cloud-init-output.log`
Review Apache logs	`sudo journalctl -u httpd --no-pager -n 50`

7. Fixing the "It works!" Default Page by Hand

1. Open the file: `sudo nano /var/www/html/index.html`
2. If the file is blank, paste the complete HTML (terminal paste is often Ctrl+Shift+V, or right-click > Paste).
3. If old code exists, remove it first. Paste only the HTML, not Markdown fences.
4. Save with Ctrl+O, press Enter to confirm, then exit with Ctrl+X.
5. Restart Apache: `sudo systemctl restart httpd`
6. Verify: `sudo head -5 /var/www/html/index.html`

8. Part E: Common Problems

Problem	What to check
Browser timeout	Security group allows HTTP TCP/80 from 0.0.0.0/0; the instance has a public IPv4 address; subnet routing is configured.
Connection refused	Run `sudo systemctl status httpd` and restart Apache if needed.
Still shows "It works!"	Check /var/www/html/index.html and the Apache DocumentRoot. Make sure the new code was saved to the correct file.
User Data did not run	Review /var/log/cloud-init-output.log. User Data runs on first boot only; editing it later does not rerun it.
Fonts look different	External Google Fonts need internet access; the page falls back to system fonts.
SSH does not connect	SSH needs a key pair and port 22 open to your IP. Without a key pair, use Systems Manager or another supported method.

9. Part F: Security and Cost Notes

- HTTP port 80 open to everyone makes the demo public. Never put passwords, secrets or private data on the page.
- Restrict SSH port 22 to **My IP**.
- This demo uses HTTP only. HTTPS needs a certificate and extra configuration.
- Terminate the instance when finished, then check EBS volumes, snapshots and Elastic IP addresses, since some can keep incurring charges.

10. Interview Explanation

> "I deployed a custom Apache web server on an Amazon EC2 instance running Amazon Linux 2023 in the US East (N. Virginia) Region. I automated Apache installation and webpage creation using EC2 User Data, configured a security group for HTTP traffic on port 80, and verified the site through the instance's public IPv4 address. I also practiced troubleshooting the Apache service, document root, and cloud-init logs."

11. Final Checklist

[ ]	EC2 instance launched with Amazon Linux 2023 in US East (N. Virginia).
[ ]	Public IPv4 address available.
[ ]	Security group allows inbound HTTP on port 80.
[ ]	User Data script pasted without Markdown fences.
[ ]	Apache service is active.
[ ]	The correct /var/www/html/index.html is present.
[ ]	The custom website opens from the browser.
[ ]	Resources are terminated or cleaned up when no longer needed.

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